Industrial Control
25791

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Assignment 3

Due Date: 1405/03/18

Note: Bonus problems are indicated in ***purple***.
Note: Remember to upload your Simulink/MATLAB files alongside your main file.

Hint: If you want to use Simulink environment, it's recommended to use a `fixed step-size` (e.g., $T = 0.001$) in all simulations of this assignment.

1 Investigating Performance of ADRC through Simulation

In this problem we want to become acquainted with linear (discrete and continuous) versions of ADRC scheme and carefully investigate its pros and cons. Do all parts of this question with respect to reference [1] through regenerating its results. For each part provide the similar figures in your PDF file.

a) **Continuous first order LADRC**: Do the following parts using MATLAB **Simulink** environment

1) Regenerate results shown in `Fig.3` for $K = 1, 0.1, 10$ and $T = 1, 0.1, 10$.
2) Regenerate results shown in `Fig.4` for $T = 1, 0.1, 10$.
3) Regenerate results shown in `Fig.6` parts `c` and `d` for $T = 1, 0.1, 10$.
4) Regenerate results shown in `Fig.7` for $T_{dead} = 0.1$. Also check for the case that T_{dead}^{ESO} is set to 0.1 instead of 0.05. Also check for the case that s^{ESO} is set to 5 instead of 2. Is there any difference if we simulate delay at input or output of process?
5) Regenerate results shown in `Fig.8` for $T_2 = 0.05$. Also check for the case that s^{ESO} is set to 5 instead of 2.
6) Regenerate results shown in `Fig.9` for $K = 1, 0.2, 5$ and $T = 1, 0.2, 5$.
7) Regenerate results shown in `Fig.10`. Also investigate effects of k_{ob} on performance of disturbance rejection.

b) **Continuous second order LADRC**: Do the following parts using MATLAB **Simulink** environment

1) Regenerate results shown in `Fig.11` for $K = 1, 0.1, 5$, $D = 1, 0.1, 10$ and $T = 1, 0.5, 3.5$.
2) Regenerate results shown in `Fig.12` for $T = 1, 0.5, 3.5$.
3) Regenerate results shown in `Fig.16` for $K = 1, 0.2, 5$, $D = 1, 0.2, 5$ and $T = 1, 0.5, 3.5$.
4) Regenerate results shown in `Fig.17`. Also investigate effects of k_{ob} on performance of disturbance rejection.

c) **Discrete first order LADRC**: Do the following parts through MATLAB **mfile** coding. You might use these codes to implement LADRC on microprocessors in future.

1) Regenerate results shown in `Fig.18`.
2) Regenerate results shown in `Fig.20`.

Hint: There may be slight discrepancies (specially high frequency oscillations) between your results and those in paper due to choice of the sampling value and the discritization method.

2 Investigating Stability of ADRC through Simulation

You have learned (at least let hope), how to investigate closed-loop stability of a first or second order LTI system under a linear active disturbance rejection control scheme. In this problem we aim to confirm obtained theoretical results through simulation. To this end, let consider following unstable system introduced in Eq. (73) of ADRC lecture

$$G_3(s) = \frac{1}{s^2 - 100s + 160}$$

a) Using simulation analysis, show that setting $T_{settle} = 0.1$ and $k_{ob} = 100$, the closed-loop system is stable for interval $0.07 < k_e < 8.4$.

Hint: If you are using Simulink, you might need to decrease the simulation step time dt for this specific example, since the closed-loop system is fast.

b) Using simulation analysis, show that setting $T_{settle} = 1$ and $k_{ob} = 10$, there is no value of k_e that stabilizes closed-loop system.

c) Setting $T_{settle} = 1$, the stable interval of $G_3(s)$ in $k_e - k_{ob}$ plane is depicted in Fig. 1 (which is simply obtained by numerically evaluating eigenvalues of A in (69) for various conditions). Choose 2 random points inside and 2 random points outside the stable region and check either closed-loop system is stable or not.
Is there any interesting point upon this figure?

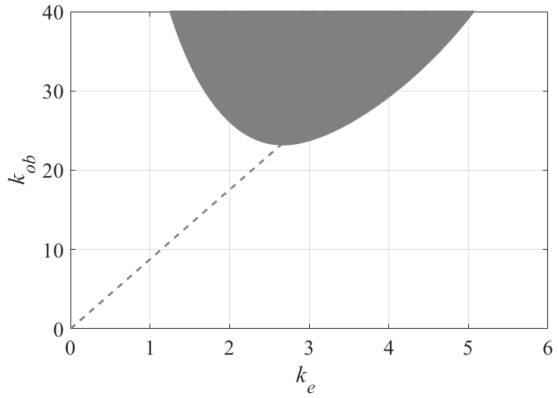


Figure 1: Stable interval of LADRC in $k_e - k_{ob}$ plane.

d) The stability analysis of second-order LADRC applied to an uncertain second-order system, for the special case $b = 1$ and $c = 1$, is given in class. Generalize this analysis by deriving the results for arbitrary parameters b and c.

e) The stability analysis of second-order LADRC applied to an uncertain second-order system is given in class. In practice, high-order systems are often modeled with simple low-order dynamics. Analyze the stability of a first-order LADRC applied to a second-order system. Validate your obtained theoretical result through simulation analysis for an arbitrary system by investigating the boundary stability conditions.

3 Calculating Derivatives of Reference Signal to be Utilized in ADRC Scheme

In this problem we aim to first calculate differentiation of a noisy reference signal $v(t)$. Then, we aim to investigate performance enhancement due to use of reference signal differentiation in control. To this end, a noisy `RefSignal` with `RefTime` vectors possessing $40s$ duration are provided for you.

a) Calculate the filtered reference signal denoted by $v_1(t)$ and its derivative denoted by $v_2(t)$ using following methods and compare the results.

I) Backward-Euler followed by low-pass filtering

II) A linear tracking differentiator

III) A nonlinear tracking differentiator

IV) An arbitrary nonlinear tracking differentiator from literature

b) For system $G_{DC}(s)$

$$G_{DC}(s) = \frac{\omega(s)}{V(s)} = \frac{12.5}{0.15s + 1} \ \frac{deg}{s.V}$$

which represents a DC motor dynamic, design a suitable LADRC controller and plot the following responses **with** and **without** using the reference differentiation (i.e., setpoint weighting) in control

I) Response to unit step reference signal at absence and presence of a reasonable angular velocity measurement noise.

II) Response to sinusoidal reference signal with unit amplitude and $1Hz$ frequency at absence and presence of a reasonable angular velocity measurement noise.

III) Response to sinusoidal reference signal with unit amplitude and $10Hz$ frequency absence and presence of a reasonable angular velocity measurement noise. Discuss upon the obtained results in part `I` to part `III`.

IV) Response to `RefSignal` utilized in part `a`. Vary tuning parameters b and c, and discuss their influence on the tracking performance.

Hint: In part `b` use an appropriate differentiation method based on your deduction from part `a`. You might want to use different parameter setting in various questions of part `b`.

References

[1] G. Herbst, "A simulative study on active disturbance rejection control (ADRC) as a control tool for practitioners," *Electronics*, vol. 2, no. 3, pp. 246–279, 2013.